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Enhancing Horizontal Drilling Efficiency and Reducing Non-Productive Time in Hot Shale Reservoir

H. Zhang, A. Alexeyenko, and A. B. Albakhrji, Saudi Aramco, Dhahran, Eastern Region, Saudi Arabia

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Abstract

Drilling horizontal wells through weak and thermally sensitive shale formations requires addressing wellbore instability challenges, including equivalent circulating density (ECD) fluctuations and mechanical disturbances. This study focuses on engineering solutions that mitigate these issues in an unconventional hot shale reservoir, improving drilling efficiency and reducing non-productive time.

The study employed a multidisciplinary approach combining geological analysis with engineering practices to address wellbore instability. Formation properties, such as mineralogy, natural fractures, bedding plane weakness, and mechanical behavior, were thoroughly evaluated to identify wellbore instability risks. To mitigate lateral wellbore instability, the drilling fluid design, managed pressure drilling (MPD) techniques, and bottom-hole assembly (BHA) optimization were assessed and refined. The enhanced hole cleaning protocols and effective cuttings management strategies, supported by real-time monitoring and optimized connection and tripping practices were implemented to further reduce risks of hole pack-off and stuck pipe and ensure an efficient drilling operation.

The implemented measures significantly improved drilling efficiency and addressed wellbore instability challenges in the hot shale formation. Non-productive time (NPT) was reduced from 40.1% to 12.8%, reflecting more efficient operations. The use of flat rheology oil-based mud (OBM) generated low and stable ECD, which minimized stress on wellbore and reducing cavings, lost circulation, and pipe sticking events. The motorized rotary steerable system (RSS) BHA reduced mechanical vibrations, enhancing stability during drilling. MPD ensured precise pressure control and minimizing ECD fluctuation, further mitigating wellbore instability risks. The implementation of enhanced hole cleaning protocols and optimized connection and tripping practices improved cuttings management, reducing the incidence of hole packing and ensuring consistent wellbore conditions for subsequent logging and completion activities.

These engineering interventions not only improved operational safety but also provided a replicable framework for horizontal drilling in thermally sensitive shale formations. The integration of advanced monitoring systems and optimized practices demonstrated that wellbore stability could be effectively managed, even in challenging geological conditions. The study highlights the importance of a multidisciplinary approach to address the complex mechanical and geomechanical challenges posed by unconventional formations, offering valuable insights for future drilling campaigns in similar reservoirs.

This outcome validates the strategies' effectiveness and underscores the need for continuous evaluation and adaptation of drilling techniques to maximize efficiency and minimize operational risks.

This study advances the understanding of horizontal drilling in hot shale formations. Areas for further exploration include continuous circulation systems (CCS) and casing centralizer optimization, offering potential improvements in operational safety and efficiency.

Introduction

Well X, a re-entry, was the first horizontal appraisal well across unconventional hot shale reservoir. The well was re-entry in 2022 by plugging back and sidetracking in 7" liner with cased hole whipstock. 5-7/8" curve section was drilled and landed the well at hot shale reservoir with 90-degree inclination and 350-degree azimuth. A standalone RSS) and conventional OBM with MPD drilled 5-7/8" lateral at an average 49.0 ft/hr rate of penetration (ROP). After 2,950 ft of lateral footage was achieved, the drilling operation was halted due to hole instability issues, which manifested as a sudden increase in standpipe pressure and subsequent fluctuations. Despite multiple attempts to restart the pump, the required flow rate could not be achieved, leading to pulling out of hole. Two attempts to run back to bottom and establish circulation were unsuccessful due to a tight hole and packing-off. A slick BHA with underreamer was run to enlarge the lateral hole, however the hole was packed off and string got stuck. Attempts to free the stuck string were unsuccessful. A decision was made to plug back the lateral (L0-1 in [Figure 1](#)) and sidetrack.

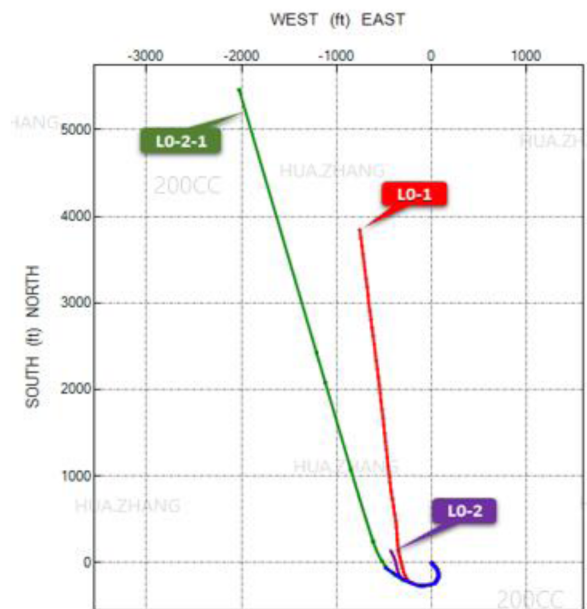


Figure 1—Well X's profile

The new 5-7/8" curve (L0-2 in [Figure 1](#)) was drilled, but unfortunately, it experienced a jar twist-off incident. fishing operation was subsequently performed but it was unsuccessful. As a result, the L0-2 curve was abandoned and a second sidetrack was initiated. The second 5-7/8" lateral (L0-2-1 in [Figure 1](#)) in same target was successfully drilled with motorized RSS and OBM to the planned well's total depth (TD). The instant ROP was controlled at 25-30 ft/hr to avoid the hole cleaning issue. In addition, thru-bit logging was performed. During the running of the 4-1/2" production casing, it had to be pushed with 50-80 klbs slack off weight from the middle point of the lateral to the bottom. Tabular cavings were observed while running the casing ([Figure 2](#)). The casing was successfully landed to bottom and the cementing job was performed. The well X took 133.5 days to complete, with 54.7 days (40.1% of the total well duration) of NPT due to well instability events ([Figure 3](#)).



Figure 2—Tabular and angular cavings

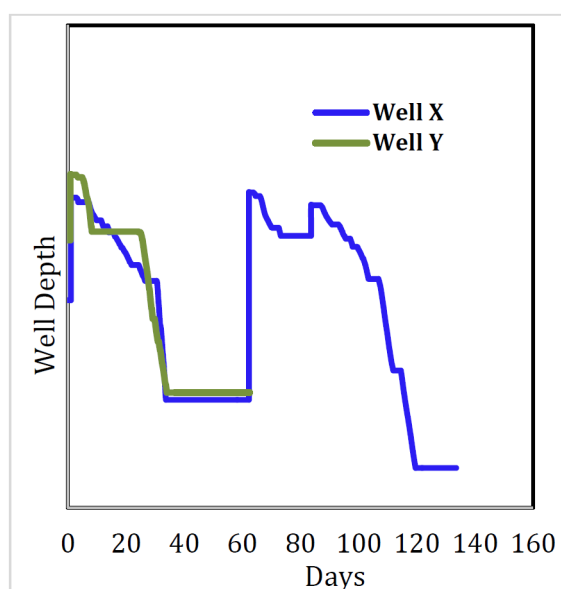


Figure 3—Time Vs depth curve comparison between well X and Well Y.

Well Y was the second horizontal well drilled in the hot shale reservoir in 2023. After the 8-3/8" vertical pilot hole was drilled and evaluated, a sidetrack was performed with OH whipstock. The 8-3/8" curve section was drilled and landed in the hot shale reservoir target with 89.5-degree inclination and 337- degree azimuth. The 7" liner was run and cemented in the curve section.

The 5-7/8" lateral section was drilled using a motorized RSS, and flat rheology OBM and MPD. Approximately halfway through the lateral, hole instability was encountered, resulting in the drill string becoming stuck and then subsequently was free. The dynamic down hole losses were introduced by surge during freeing the string. Then the drilling team experienced a well control event, the well was killed and dynamic losses were detected. Surface cuttings analysis revealed 30-40% tabular cavings. A short trip was performed, during which several tight spots were identified, but full flow rates were successfully established with full return at the bottom of the hole. Drilling resumed and continued to the well TD. Logging

operations were successfully performed using the thru-bit logging tool without incident. Two sedimentary faults associated with conductive fractures in the lateral were detected by a post-drilling log.

The initial attempt to run the 4-1/2" casing string was unsuccessful due to tight spots and hole packing off. A 5-7/8" slick clean out assembly was used to effectively clean the hole and remove cavings. Upon running the casing string without centralizers, it was successfully landed at bottom of hole. The cementing job was performed, but hole packing off was observed during displacement. The cementing operation was completed by displacing the cement at a reduced pump rate. Well Y was completed in 62.4 days, with 8.0 days (12.8% of the total well duration) of NPT due to well instability events (see Figure 3). In comparison to well X, the performance on well Y marks a significant improvement with NPT associated to wellbore instability decreasing from 40.1% to 12.8%. The substantial enhancement in drilling efficiency and mitigation of wellbore instability challenges in Well Y can be directly attributed to the effective implementation of insights gleaned from comprehensive formation property analysis and drilling engineering studies.

Formation Properties

Hot shale formations present unique drilling challenges due to complex geology structure, mechanical behavior. To better understand these challenges, the project team conducted in-depth geological analysis including mineralogy, natural fractures, bedding plane weaknesses, and mechanical behaviors.

Mineralogy and Reactive

X-Ray diffraction test on the core sample from the well Y vertical pilot hole shows that the hot shale contents include high quartz (44.9%) and high clay (35.8%), low calcite (3.5%) and low dolomite (2.2%). The high quartz content and competent shale suggests the presence of brittle shale, which is susceptible to fluid invasion into bedding plane layers through capillary action which could result in borehole instability, as reported by Hoxha and Rabe (2021). Laboratory linear swell meter testing was also conducted on a hot shale core sample immersed in calcium chloride brine with a salinity of 250,000 ppm for a duration of 24 hours. The test results revealed a maximum core swell of only 1.6% (Figure 4), indicating a negligible swelling effect. This suggests that the shale core sample exhibited minimal reactivity to the brine, with no significant expansion or degradation observed over the test period.



Figure 4—Linear swell meter testing results on a hot shale core sample.

Fractures

Examination of the core sample from Well Y revealed the presence of both induced and natural fractures (Figure 5). The primary mechanism of interaction between the shale and fluids was found to be fracturing and delamination along the bedding planes, accompanied by the enlargement of pre-existing fractures. Notably, the fracture propagation directions were predominantly parallel to the bedding plane, with a

preferred orientation of approximately 335° in the lateral direction (Figure 6). This suggests that the fractures tended to propagate in a direction consistent with the in-situ stress field, which is likely influenced by the bedding plane orientation.



Figure 5—Core Sample from well Y vertical pilot hole.

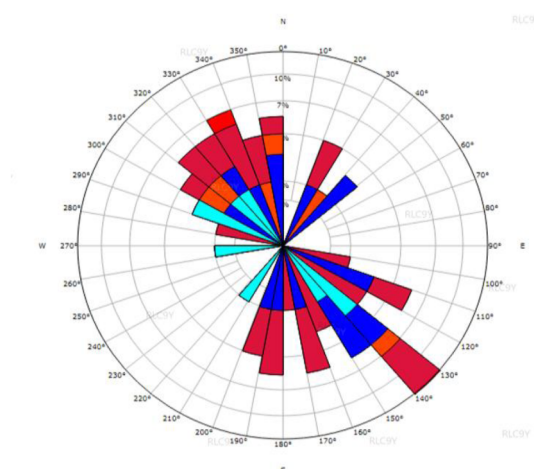


Figure 6—Fracture propagation direction was interpreted from log data in well Y's lateral.

Geomechanical Findings

Geomechanical characterization of hot shale reservoirs is essential for optimizing drilling and production operations. Our geomechanical team performed a comprehensive analysis of the reservoir's properties, including laboratory tests and log reviews, to gain a deeper understanding of the rock's behaviors. The results of the rock mechanics test (Table 1), which included measurements of the core samples from Well X and Well Y, indicate that the rocks are brittle and exhibit low ductility. The low anisotropy concludes the hot shale rock has weak bedding planes and is naturally fractured. These findings emphasize the importance of minimizing ECD fluctuations and mechanical disturbances during drilling operations to avoid hole instability and ensure safe and efficient drilling.

Table 1—Geomechanical findings

Findings	Well X	Source	Well Y	Source
Brittle	Yes	Rock mechanics tests, Cavings during drilling and fracking	Yes	Rock mechanics tests and cavings
Ductile	No	Image log/ caliper	No	Rock mechanics tests and image log/caliper
Swelling	NA		No	Lab tests
Pore pressure	Medium to high	Gas indicator and offset wells	Medium to high	Gas indicator and offset wells
Strength	UCSh:7 000 psi, UCSv:14000 psi	Mechanical rock test	UCSh: 5000-7000 psi, UCSv: 15000-17000 psi	Mechanical rock test
Stresses	Normal stress regime	Wellbore stability inversion	Normal stress regime	Wellbore stability inversion
Anisotropic	Medium Anisotropic	Eh/Ev (1.8-2.3)	Bedding and fracture with low stress anisotropy	Eh/Ev (1.8-2.3)
Hydraulic Frac Complexity	Yes	Pending Analysis (cavings while fracking)	Yes	Yes (Bedding/ Fractures) low stress anisotropy
Origin of instabilities	Yes	Hydraulic Fluctuation	Yes	Discontinuities, Hydraulic Fluctuation

Drilling Engineering Studies and Execution

Flat Rheology Oil-Based Mud

The well X lateral 0-1 was drilled with conventional OBM. After experiencing the wellbore instability issue in lateral 0-1, the new lateral 0-2-1 was drilled with optimized conventional OBM which treated the existing fluid with 50% of Micromax and 50% of Barite as weighting agents to reduce the rheology and minimize the ECD. After gaining a thorough understanding of the hot shale formation properties, the drilling team, including the mud specialist, designed a drilling fluid program for the next well Y, with a focus on minimizing ECD and inversion. The flat rheology OBM was selected as the optimal choice due to its consistent viscosity and rheological properties over a wide range of temperatures and pressures, making it particularly suitable for drilling the hot shale lateral wells where wellbore stability is a major concern. Compared to conventional OBM, the flat rheology OBM offers several advantages, including lower ECD, lower gel break, and lower pump pressure, which collectively contribute to enhanced wellbore stability and reduced drilling risks. The strategic use of polymers and asphaltic materials to counteract the effects of lignite materials resulted in an effective reduction in pore pressure transmission, thereby minimizing fluid invasion and controlling shale platelets.

As illustrated in [Table 2](#), the flat rheology OBM utilized in Well Y exhibited the lowest rheology and ECD, as well as the lowest high-temperature high-pressure (HTHP) fluid losses, when compared to the conventional OBM used in both laterals of Well Y, this data suggest that it was effective in achieving this goal of minimizing the ECD and inversion.

Table 2—Mud properties and performance comparison

Lateral Name	Well X_ L0-1	Well X_L0-2-1	Well Y lateral
Mud type	Conventional OBM	Conventional OBM Micromax/Barite=50/50	Flat rheology OBM
MPD	Yes	NO	Yes
Oil/Water	80/20	90/10	80/20
Mud Weight (pcf)	95	107	98
Plastic Viscosity (cP)	30	24	27
Yield Point (lbf/100ft ²)	20	16	15
Low Shear Yield Point (lbf/100ft ²)	9	7	5
6 RPM (θ)	11	9	7
HTHP Fluid Loss (cc/30min @300°F)	2.2	2.8	2.0
ECD at casing shoe (pcf)	104.6	112.1	101.3
ECD at TD (pcf)	109.4	117.3	105.4
Pump pressure (psi)	4300 psi / 250 gpm	4300 psi/ 240 gpm psi	4200 psi / 250 gpm
BHA Type	RSS	Motorized RSS	Motorized RSS

On-site Hydraulic Simulation and Hole Cleaning Modeling

On-site hydraulic simulation and hole cleaning modeling allow for data-driven decision making, enabling drilling teams to optimize hydraulic parameters and ROP to ensure efficient hole cleaning, prevent hole packing-off and minimize the risk of hole instability.

A valuable lesson was learned from the drilling of well X, which experienced the hole cleaning issue while drilling with an average 49 ft/hr high ROP and without on-site hole cleaning modeling verification. This experience informed the drilling of well Y, where the lateral was drilled by controlling ROP at 30 ft/hr and the Virtual Hydraulics software was used to perform real-time hole cleaning modeling while drilling the lateral section, resulting in improved drilling efficiency and reduced risks. As per the simulation the ROP needs to be controlled without exceeding 40 ft/hr ROP to ensure cutting do not accumulate (see [Figure 7](#)). The hole cleaning model also provides cutting bed size, concentration and location (see [Figure 8](#)). The model suggests a constant cutting bed of 0.08 inches was observed throughout the horizontal open hole section, with a cutting concentration of 0.3%. This concentration was maintained within an acceptable range while controlling the ROP at 30 ft/hr.

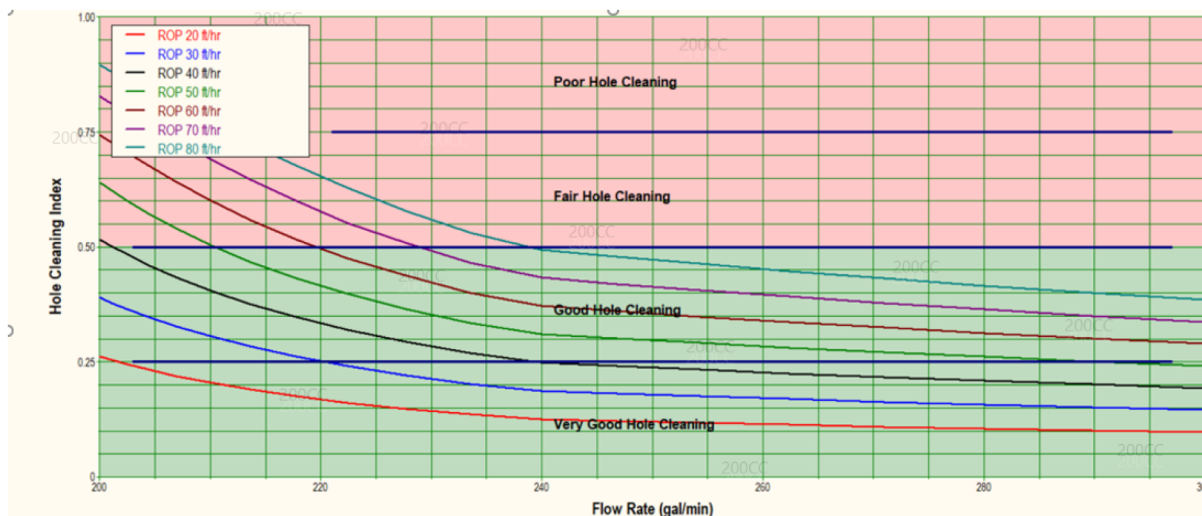


Figure 7—Hole cleaning index from Virtual Hydraulics software, hole cleaning in very good range at ROP of 20-40 ft/hr at 230-240 gpm flow rate.



Figure 8—Hole cleaning model shows constant cutting bed of (0.08 inch) all over the horizontal open hole section and cutting concentration of 0.3% with applied ROP of 30 ft/hr.

BHA and Parameters

The wellbore instability is attributed to high drill string vibrations caused by high surface revolutions per minute (RPM) and weight on bit (WOB) as well as high mud flow rates [Khan et al. \(2021\)](#). This phenomenon is consistent with the findings of [Sun et al. \(2019\)](#) who reported a similar behavior of increase in wellbore enlargements due to high WOB and RPM in large-scale laboratory experiments on shale blocks subjected to anisotropic horizontal stresses.

The 5-7/8-inch lateral section of Well-X_0-1 was drilled using a standalone RSS BHA with high WOB (18-20 klbf) and high RPM at 120. During drilling, all three vibration types - axial (longitudinal), transverse (lateral) and torsional (stick slip) were detected by RSS. The 5-7/8-inch lateral of well Y was drilled using a motorized RSS which featured a more optimized design BHA together with lower surface RPM (40-60) and WOB (5 and 12 klbf) while maintaining same flow rate. No axial or transverse vibrations were detected while the torsional vibration (stick slip) was lower compared to well X ([Figure 9](#)). This observation suggests the motorized RSS operated with lower RPM and lower WOB resulting in lower vibrations, can effectively reducing drill string mechanical disturbance and improving hot shale lateral wellbore stability.

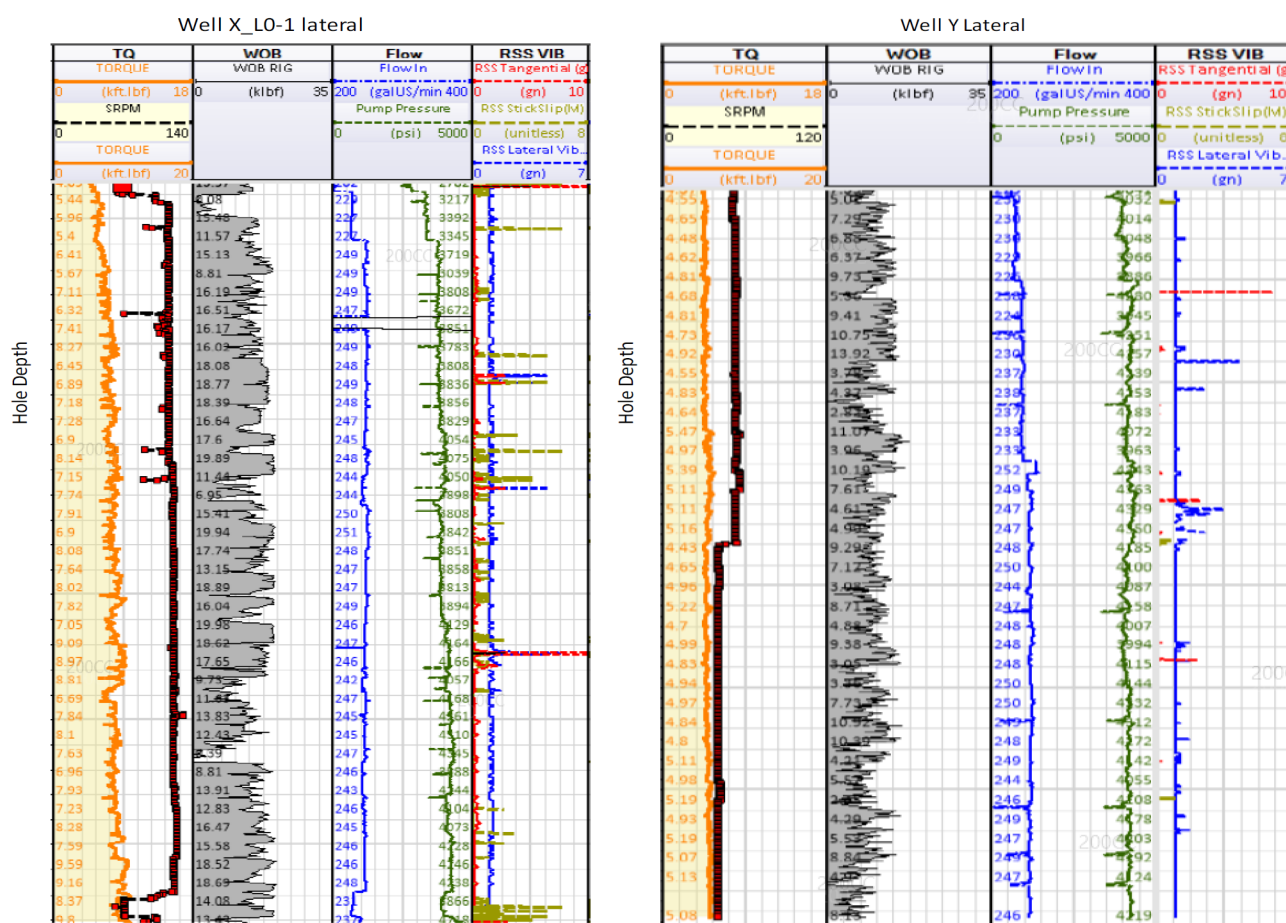


Figure 9—Comparison of drilling parameters and vibrations while drilling lateral in well X and well Y.

MPD Strategy

Change in ECD causes the wellbore to become unstable, leading to collapse or fracturing. MPD is a drilling technique used to minimize the ECD fluctuation.

In MPD operations, the anchor point is a critical parameter that refers to the depth in the wellbore where a target ECD value is maintained constant in both static and dynamic situations. However, outside of the anchor point, the ECD in the wellbore can change when switching between drilling and making connection conditions. Minimizing ECD fluctuations is a key objective of MPD operations while drilling in hot shale reservoirs. To achieve this goal, a tailored MPD anchor point strategy was implemented for drilling the Well Y_L0-1 lateral. The strategy involved setting the anchor point at the bit depth while drilling from the landing point to the middle of the lateral, and then change the anchor point to the middle point of the lateral while drilling rest interval to well TD.

Figure 10 shows the ECD profile while drilling the 5-7/8- inch lateral of well Y from landing point to middle point of lateral. The MPD anchor point was set at the bit depth and the ECD at the shoe was 104 pcf while drilling and ECD was 106 pcf while making connection. The ECD at bottom of hole remained constant while that along the wellbore changed in the range 0-2.0 pcf when making connections. In contrast, as shown in Figure 11, the MPD anchor point was set at the middle point of the lateral, the ECD at the middle point of lateral was 105.5 pcf while drilling and 108 pcf while making connection. The ECD at the bottom of the hole was 109.5 pcf while drilling and 108 pcf while making connection. As a result, the ECD change along the wellbore was in the range of 1.5-2.5 pcf.

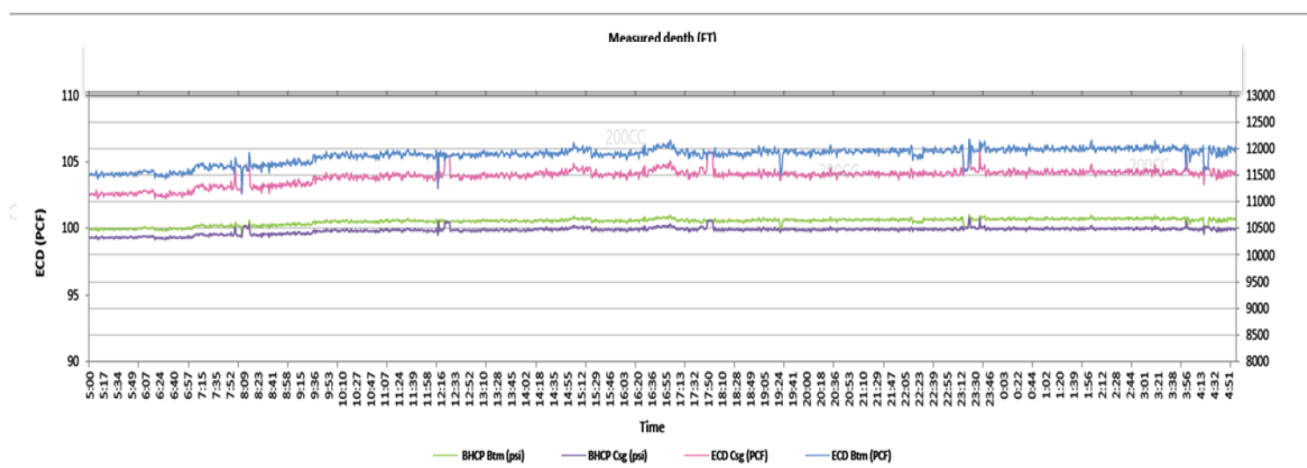


Figure 10—MPD ECD profile while drilling from landing point to middle point of lateral in well Y.

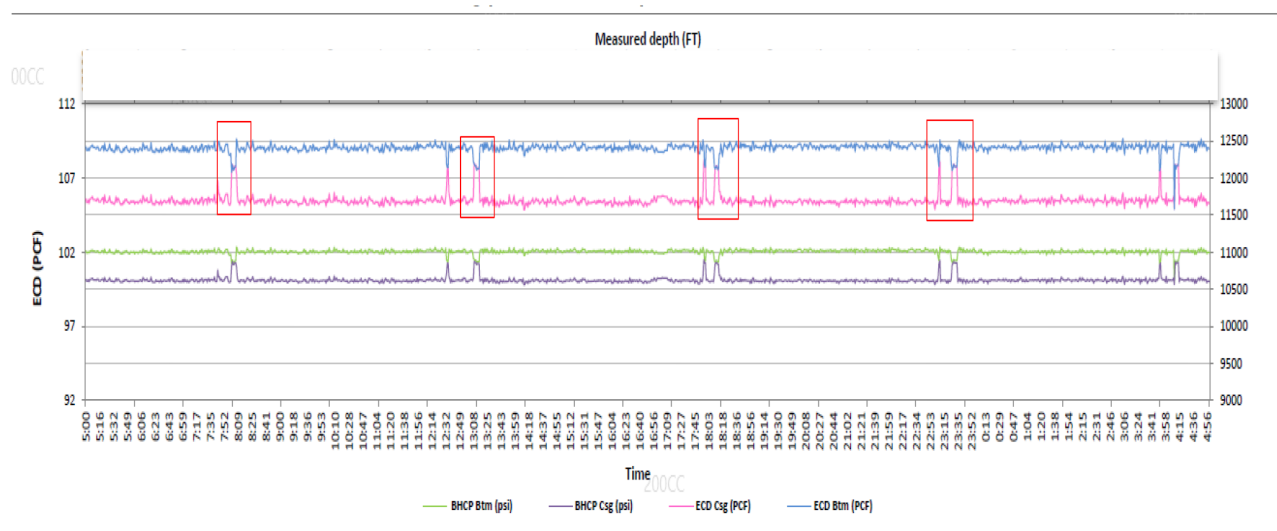


Figure 11—MPD ECD profile while drilling from middle point of lateral to TD in well Y.

The pressure while drilling (PWD) sensor was run with a directional tool in well Y to provide bit depth ECD during drilling and also equivalent static density (ESD) while making connection. The difference between the ECD and ESD reflects the bottom hole pressure fluctuation. The measured ECD and ESD difference was in a range 1.5- 3.6 pcf. (Figure 12). Highly correlated with MPD system output, it demonstrated the accuracy and reliability of the optimized anchor point setting strategy employed on the MPD system.

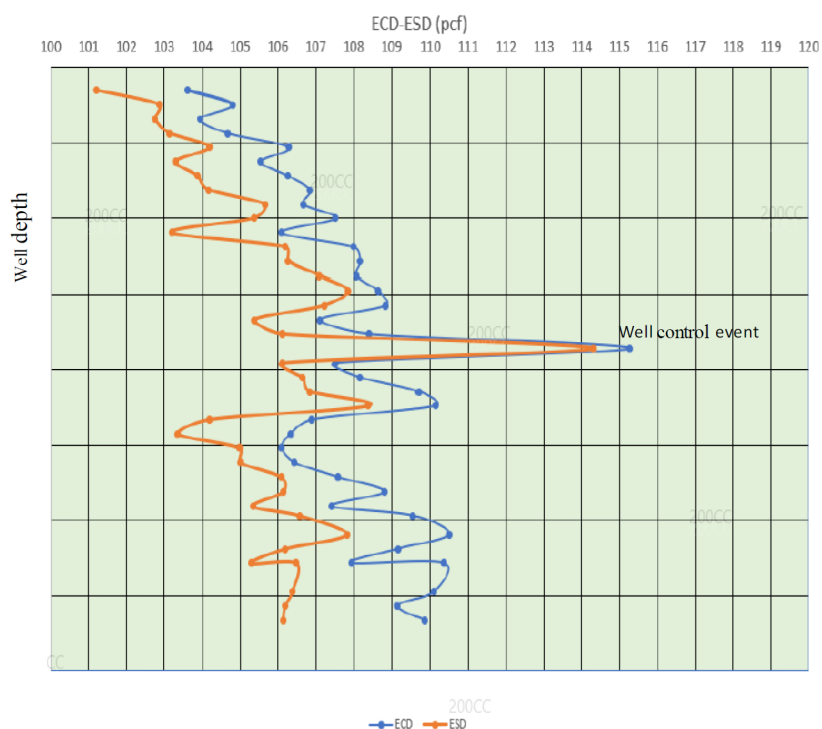


Figure 12—ECD and ESD measurement with PWD in well Y.

Drilling Practices

Connection tripping Practices. During normal connection operations, erratic torque and packing off issues were encountered. Investigation revealed that these problems were caused by the accumulation of cuttings and cavings in the 5-7/8" small hole, resulting from flow area restrictions around the BHA. To address this issue, an optimized connection practice was developed to provide sufficient circulation time for removing cuttings and cavings around the BHA, and minimize the back reaming to avoid mechanical disturbance. The new procedure consists of the following steps:

- Drill off for 5 minutes to create a clean hole environment.
- Back ream slowly at 20 RPM for 20ft to help remove any accumulated debris.
- Circulate and rest the hole for 1 minute prior to recycling the pump for taking survey.
- Recycle the pump and take survey.
- Ream down slowly at 20 RPM to the bottom.
- Circulate for 8 minutes at the bottom of the hole to ensure all cuttings and cavings are removed.
- Pick bit off the bottom as high as possible for connection, stage down the pump, and make the connection quickly.
- Stage up the pump, start rotation, and resume drilling operations.

The implementation of this optimized connection practice successfully mitigated hole packing off issues during subsequent connection operations.

Tripping Practice. Both well X and well Y encountered difficulties during tripping out operations, with tight hole and packing off issues indicating that hole cleaning remained a concern due to the presence of a small amount of cavings. To address this issue, the operations team implemented a staged circulation

procedure at specific intervals along the lateral section during tripping out. The staged circulation points were set at $\frac{1}{4}$ of the lateral depth, $\frac{1}{2}$ of the lateral depth, $\frac{3}{4}$ of the lateral depth and at the shoe.

Areas for Further Study

The drilling engineering team has identified several areas for further study, including the implementation of CCS and optimization of casing centralizers. These initiatives have the potential to yield significant improvements in operational safety and efficiency for drilling hot shale lateral.

While MPD can minimize ECD fluctuations, it is not able to completely eliminate hydraulic disturbances to the wellbore during connections. However, the CCS maintains continuous circulation and a consistent downhole ECD during connections. Combining MPD with CCS could provide a comprehensive solution. The synergistic benefits of combining CCS and MPD include:

- Maintaining a consistent bottomhole pressure (BHP), mitigate ECD fluctuations and reduced risk of wellbore instability.
- Improved hole cleaning and reduced risk of pack-offs.
- Enhanced downhole temperature control.
- Potential for increased drilling efficiency and reduced NPT.

Maintaining good circulation is crucial for the production casing running and cementing operations. A higher flow rate centralizer plays a significant role in preventing hole packing off. In this regard, the helix centralizer offers a distinct advantage over conventional solid centralizers, such as spiral solid centralizer and single piece bow centralizer, providing the highest flow area ([Table 3](#)) and thereby optimizing circulation.

Table 3—Flow area comparison on typical type casing centralizers

4-1/2" casing × 5-7/8" open hole	Spiral solid centralizer	Single piece bow centralizer	Helix centralizer
Flow Area	5.63 (Sq in)	3.93 (Sq in)	9.29 (Sq In)

Conclusions

The hot shale formations exhibit complex formation properties, including brittleness, pre-existing fractures, weak bedding planes and potential faults. These properties pose significant wellbore instability challenges during the drilling of horizontal laterals in the hot shale reservoir. To address these challenges, it is essential to minimize ECD fluctuations and mechanical disturbances during drilling operations.

The implementation of flat rheology OBM and MPD minimized the fluid inversion, reduced the ECD and ECD fluctuations, while a motorized RSS reduced BHA vibrations and mechanical disturbance on wellbore to improve the hole stability. Additionally, enhanced hole cleaning protocols and real-time monitoring improved cuttings management, and dedicated connection and tripping practices reduced risk of hole pack-off and pipe stuck incidents. The successful implementation of these measures resulted in significant improvements in drilling efficiency, with NPT reduced from 40.1% to 12.8% in challenging geological conditions.

Looking ahead, areas for further study include the implementation of CCS and casing centralizer optimization, which offer potential improvements in operational safety and efficiency.

Acknowledgement

The authors wish to thank Fahad Alkhunaini and Francis Elisabeth, both of Saudi Aramco, for providing [Fig. 5/](#)[Fig. 6](#), and [Table 1](#), respectively.

Abbreviations

NPT	– Non-productive time
MPD	– Managed pressure drilling
BHA	– Bottom hole assembly
OBM	– Oil base mud
HTHP	– High-temperature high-pressure
ECD	– Equivalent circulating density
ESD	– Equivalent static density
ROP	– Rate of penetration
WOB	– Weight on bit
RPM	– Revolutions per minute
PWD	– Pressure while drilling
CCS	– Continuous circulation system

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